

Research Article

Applying Facebook Prophet to forecast the passenger flow in a seaport

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Abstract

In this study, a robust model for forecasting passenger arrivals at the Vlora seaport was developed using the Facebook Prophet method. The analysis was based on monthly data for the period 2014 - 2024, with the aim of filling the gap in the Albanian literature, in which advanced forecasting approaches for seaports are lacking. Through a comparative analysis with other forecasting methods (ARIMA, Holt-Winters, Monte Carlo, and artificial neural networks), it was found that the FB Prophet model explains 96.7 % of the data variance with a MAPE of 24.95 %, suggesting high forecasting accuracy. Also, the integration of external factors (e.g., exchange rate and GDP) showed a limited impact, highlighting the importance of historical data for this approach. These results provide a basis for managerial suggestions for the Vlora seaport, which include optimizing service lines, determining the type and size of ships, deciding on pricing policies, and organizing staff, so that the port can be better integrated into the international transport network. The study has particular value and great importance, as it fills a gap in the Albanian literature, provides an advanced methodological framework and contributes to economic development and regional integration, supporting port authorities and operators in long-term infrastructure planning and in improving the quality of passenger services.

1. Introduction

Located in the southwest of Albania, Vlora is a pearl in terms of its beauty. Vlora is not only a beautiful city, but is also a very important city for the history, culture, and economy of the country. It contributes about 10 % to the Albanian GDP. Vlora is the city where Albania's independence was proclaimed, the arena of wars for freedom and Albanian resistance, the birthplace of the father of the nation, Ismail Qemali, the city where Skënderbeu was married, a natural studio for painters, a home to artists, and a bridge connecting Albania with the world. The Declaration of Independence in Vlora, its geographical position, and its important connections through the seaport have earned it several titles: "city of the flag", the "city with two seas", the "pearl of Albanian tourism", and "the southern port of the Adriatic". During the years of the dictatorship in Albania, on November 28 and 29 each year, Vlora was symbolically referred to as the capital of Albania, in recognition of its historical significance as the city where independence was declared.

With an area of 2,706 km², the city stretches from north to south with a length of 19 km and an east-west width of 16 km (Gjika, 2024). According to the INSTAT, the population of Vlora in 2024 is around 124,000 people (INSTAT, 2024). Vlora is one of the biggest cities of Albania and its population ranks it 4th in population size (Population HUB, 2024). It borders other cities like Fier, Mallakastër, Tepelenë, Sarandë, and Gjirokastër, and on the west, it is surrounded by the Adriatic and

Ionian Seas. This makes Vlora a strategic link between Europe and the Mediterranean. It is 72 km from Otranto (Italy), and 77 miles from Greece (Bashkia, 2025), so its port is significant in international maritime connections. In the 4th century, the port of Vlora was among the most important, which not only provided a bridge between Albania and other European countries, but also helped in the Albanian economy through the transportation of agricultural and livestock products to Trieste, Venice, Corfu, Istanbul, Izmir, Manastir, Malta, and Cyprus. The Vlora seaport is a strategic location on the transport corridor towards Europe and other countries up to Albania. In the 17th and 18th centuries, Vlora was one of the main Albanian seaports, and the bay sheltered ships from storms. Today, Vlora is one of the largest and most important seaports in Albania, after Durrës, in terms of size and capacity. It is the biggest port in the south and serves all the southern regions of Albania in commercial and tourist transportation.

Although Vlora seaport is a very important factor in economic development and regional integration, there is a lack of accurate predictive models that would assist in strategic decision-making and infrastructure investment. This paper addresses an important gap in the Albanian literature, as advanced forecasting approaches for passenger flows in Albanian ports have been absent to date. The results provide an extended methodological framework that can be used for managerial guidance in increasing the operational and strategic performance of the port and its penetration into the international transport system.

Figure 1 presents a relief map of Vlora Bay, which clearly shows the port's topographic features. This visualization is crucial for assessing the natural environment and for guiding infrastructural planning and port functionality development.

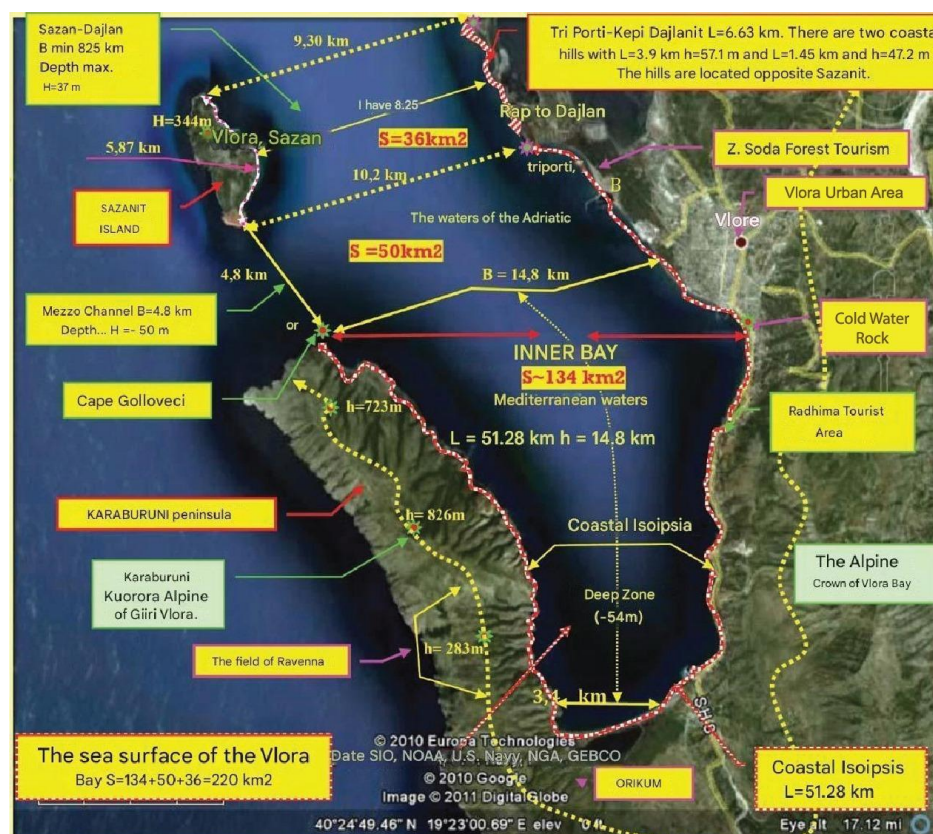


Figure 1 Relief planimetry of Vlora Bay.

Vlora seaport was built in the bay of Vlora, and has a total surface of 53,000 m² and a water basin of 5,000 m². It is an open port, with two central piers handling commercial ships and ferries, and has with a depth of approximately 11 meters. The port offers about 2,100 m² of enclosed

warehouses and 9,000 m² of storage areas for customers, which are rented out to facilitate efficient goods processing. Outside the delineated port boundaries, a safe anchorage area is provided, offering protection for ships due to the favorable geographical position of Vlora Bay. There are no water currents, tides, or strong ebbs that can affect navigation (“Sea Port” Vlorë, 2024).

In the last few years, Vlora seaport has been developed in order to increase its capacity and to make it more contemporary especially in the tourist season. Vlora seaport is a strategic infrastructure of the city, as shown in **Figure 2**.



Figure 2 Vlora seaport.

In 2021, a British-Albanian joint venture won the tender to build “Vlora Marina” on the site of the current passenger terminal for the investment and further development of the port. This luxury project comprises facilities for yacht berths, residential and hotel complexes, commercial areas, etc. Vlora Marina introduces Albania’s first authentic mixed-use urban development, featuring a world-class marina with 438 berths available year-round, premium sea-view apartments, and a prestigious network of 5-star luxury brands (Vlora Marina, 2022).

As shown in **Figure 3**, found on the official project website, the ‘Vlora Marina’ project has a modern and multifunctional design that includes luxury amenities as well as residential and commercial facilities of the new seaport of Vlora, which is being developed as a major urban maritime center to enhance regional connectivity and tourism.



Figure 3 “Vlora Marina” project.

The location, history, and development of the ‘Vlora Marine’ project are transforming Vlora seaport into a major economic and tourist hub. Vlora seaport is highly active in terms of ship handling capacity and is capable of accommodating all kinds of vessels, including commercial ships, ferries, and passenger ships. **Table 1** presents historical data on the number of incoming passengers in this port from January 2014 to December 2024, officially obtained from the Statistics Office at the Port of Vlora.

Table 1 Number of incoming passengers at Vlora seaport for the period 2014 - 2024.

Year	January	February	March	April	May	June	July	August	September	October	November	December
2014	3,908	3,093	3,660	4,062	3,157	5,446	10,855	18,191	4,673	4,103	4,133	8,393
2015	4,450	3,319	4,074	4,343	6,207	6,207	19,524	30,668	7,317	6,170	4,797	9,602
2016	5,193	4,140	4,483	3,574	5,121	6,878	16,672	21,824	6,840	5,855	4,803	10,674
2017	5,223	4,162	3,517	5,877	6,299	8,362	15,760	22,291	5,680	4,884	3,878	8,651
2018	4,304	3,350	3,610	5,045	4,816	6,699	14,391	23,251	5,875	4,986	4,365	8,160
2019	4,171	2,928	4,550	4,910	4,010	9,142	12,335	23,208	5,920	3,400	2,733	9,190
2020	4,869	3,754	980	0	128	1,741	5,144	4,582	2,066	2,328	1,709	2,174
2021	1,525	1,230	1,345	1,543	1,955	2,770	8,604	21,035	3,879	3,617	2,583	5,228
2022	2,282	2,071	2,408	3,073	3,291	4,371	21,411	34,136	6,099	4,917	3,956	9,201
2023	3,509	2,448	3,284	3,411	4,544	4,184	18,022	44,242	7,368	4,142	3,258	5,780
2024	2,242	1,918	2,440	2,462	3,701	7,281	20,885	40,919	7,158	3,380	2,891	6,341

The passenger flow at the port is not only an indicator of the travel volume but is also a significant measure of the economic and urban development of the area. A high passenger flow increases the port’s capacity to provide quality services, leading to infrastructure and tourism development investments. The reasons for ferry travel at Vlora seaport are numerous and include personal ones, such as holidays, visits to relatives, health treatments or for religious purposes and transit trips, while a significant part is related to business and professional needs. Recently, a particular growth in dental tourism has become a key factor in the increase of passengers choosing Vlora for quality medical services due to their lower costs.

Tourism arrivals in Vlora seaport are closely linked to the city’s cultural, historical, and natural attractions, thus reinforcing the port’s image and importance as a key point for regional development. Although the Port of Durres is the largest Albanian port, with high cargo capacities, the strategic position of Vlora seaport makes it a key link for central and southern Albania, offering fast connections with ferries and ships for passengers, as well as import and export operations. In comparison with regional ports, Vlora seaport, in addition to local advantages and expanded infrastructure, has a favorable geographical position as a connection between domestic and regional areas. A detailed assessment between ports would enable the discovery of opportunities for regional cooperation, diversification, and development of services and infrastructure, in accordance with national and international transport strategies.

This study aims to model the number of passengers arriving by sea at Vlora seaport for 2014 - 2024, and to forecast the number of arrivals at this port for the coming years. A reliable model of the number of passengers would help not only Vlora seaport to efficiently manage their capacities and to develop the infrastructure, but also allow the local authorities and transport operators to plan the necessary capacities to optimize the services and to offer a better experience for travelers.

The first chapter of this paper presents the study’s main topic, purpose, objectives, and importance. The second chapter describes the techniques and methods used to carry out the study.

The third chapter describes several studies on modeling and forecasting passenger flow at various ports and some applications of the FB Prophet method in various scientific works by different authors. In the last chapter, the main results of the study are briefly reflected on, and suggestions are given about how they would affect an increase in the performance of this model.

2. Materials and methods

In this study, an advanced method has been applied to model and forecast the number of passengers arriving at Vlora seaport. Although other techniques, such as ARIMA, Holt-Winters, Monte Carlo simulations, and artificial neural networks, were initially evaluated, it was found that these approaches did not capture complex data patterns as effectively as Facebook Prophet. This method, Facebook Prophet, is effective in implicating external influencing factors in the modeling and forecasting of passenger flow. The following Figure 4 clearly shows the variations occurring in specific periods. Vlora seaport has an increase in passenger flow in the summer months due to the summer holidays, making July and August the maximum values for every year. In the autumn months, the flow drops significantly to register higher values in December, corresponding to New Year's Eve.

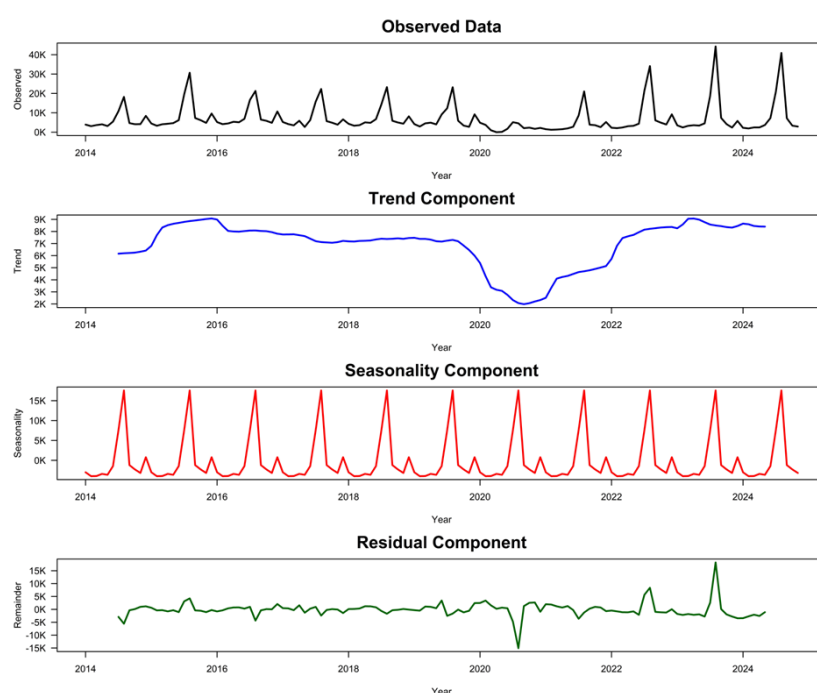


Figure 4 Seasonal and trend components.

In this study, monthly data on the number of arrival passengers at Vlora seaport from January 2014 to December 2024 was used, which was received officially from the statistics office, Vlora seaport; monthly data for the exchange rate, which was downloaded from the official website of the Bank of Albania (official exchange rate); and data on the Gross Domestic Product of Albania, which was obtained from the official website of the World Bank (World Bank Group- International Development, Poverty and Sustainability). GDP and exchange rate were integrated into the model as external regressors, helping to capture their impact on seasonal and long-term changes in passenger flows. Considering the impact of external factors, the study was structured into four scenarios to build the most effective and appropriate model for predicting passenger flows.

The Rstudio software was used for data processing and model application. Some basic packages had to be installed in advance with the code: `install.packages("ts", "prophet", "forecast", "lubridate", "plotly")`.

Historical data was analyzed and cleaned to eliminate anomalies and extreme values. For example, zero values recorded during the COVID-19 pandemic were treated as missing values and processed using the integrated capabilities of the FB Prophet model.

Eighty percent of the data was used for training, and 20 % for testing, to evaluate how the model performed for future values. This separation ensured that the model's performance was assessed on data that was not used during the training process. To evaluate the study's accuracy, the standard measurement metrics MAE, MAPE, RMSE, and R^2 were used, which showed that the model captured 96.7 % of the data variation. The results illustrate that FB Prophet is reliable for supporting strategic port capacity management.

Interactive graphs, built with the plotly package in R, allow stakeholders to dynamically explore seasonal trends and long-term changes, providing a user-friendly interface for detailed data analysis.

3. Literature review

Albania is located in the Western Balkans and contributes to the integration of European and international transport links as the shortest route between the Black Sea and the Adriatic Sea. (Mazrekaj, 2015) Mazrekaj explored the role of sustainable infrastructure and green design in Albania, which is identified as a transport route and corridor between the Adriatic Sea and the Black Sea. While the network of seaports in Albania facilitates economic development and integration of regional areas, Vlora seaport is an essential strategic element. However, to this paper's knowledge, no scientific study has been conducted to date on the modeling and forecasting of passenger arrivals at this port. This lack of Albanian scientific literature presents a unique demand to develop reliable predictive models for Vlora seaport. This can enhance strategic decision-making and better integrate this port into the European and international transport network.

International studies use advanced methods, such as time series analysis, neural networks, and artificial intelligence techniques, to address the challenges of passenger flow forecasting in seaports. Good results are obtained, and a valuable methodological framework is provided for operational management and infrastructure investments.

The study "Passenger Ship Forecasting with the SARIMA Approach (Random Case Study: Semayang Port, Balikpapan City)" (Multiningsih et al., 2022) includes a step-by-step approach to forecasting passenger flow in a seaport using the SARIMA model. In this paper, the authors examined the time series data of inbound and outbound passenger flows at Semayang port and identified seasonality using ACF and PACF functions, ensuring the stationarity of the data through necessary transformations.

After analyzing the data, the findings showed that the best model for forecasting outbound passengers at the port of Balikpapan City was SARIMA(4,1,0)(0,1,2)[12], with a MAPE of 14.05 %. The most effective model for inbound passengers was SARIMA(0,1,1)(2,1,0)[12], with MAPE values of 3.27 %, which indicates high forecast accuracy.

In another paper "Demand Forecasting Model for Airline Flights Based on Historical Passenger Flow Data" (Lundaeva et al., 2024), the authors used the Facebook Prophet method and highlighted the challenge of forecasting by including seasonal trends such as annual, monthly, and weekly trends, as well as external factors, including regional gross product, average income, and population in the regions of origin and destination. The proposed model consisted of three main stages: (i) pre-processing of historical data to identify seasonal trends and patterns, (ii) integration of other factors through a multiple regression model, and (iii) calibration of the model using a dynamic system approach to provide a detailed hourly forecast for the future period (2025 - 2029). The results were compared using RMSE, MAE, and MAPE, and it was concluded that, although the model is a good forecasting model, the average deviations of more than 30 % indicate room for improvement with hybrid or more sophisticated approaches.

This study is important to contribute to the entirety of knowledge on air demand forecasting by proposing a methodological framework for the development of integrated models. Furthermore, the findings of the study highlight the need to consider macroeconomic factors and seasonality in the

forecasting process. These factors also agree with the aims of our study on passenger flow forecasting at Vlora seaport using the Facebook Prophet model.

In the study “Forecasting Ferry Passenger Flow using Long-Short Term Memory Neural Networks” by Fesalbon (2024), the author presented a new approach based on LSTM neural networks for forecasting ferry passenger traffic. This paper used monthly data from 2016 to 2022 of Philippine ports to build and test the model, focusing on two major ports- Batangas and Mindoro. By applying the LSTM method, which is able to capture the nonlinear nature and time dependence of sequential data, the author achieved a forecast accuracy of about 72 % for the port of Batangas and 74 % for the port of Mindoro, according to the RMSE, MAE, and MAPE metrics. These results highlight the ability of LSTM networks to address the challenges of passenger flow forecasting in maritime transport, providing a strong methodological foundation that can be used to support strategic decision-making and operational improvement in the port sector.

The present article, which models and forecasts passenger flow in Vlora seaport by integrating the FB Prophet method with variables such as GDP growth rate and exchange rate, along with the impact of COVID-19 effects, presents a methodological comparison with international studies, placing it in a broader context, with the main challenge seen in developing accurate forecasting models that can support strategic decision-making and the evolution of the operational infrastructure of seaports. Thus, it contributes to the limited available literature by being the first to undertake such studies using the latest artificial intelligence techniques and by providing a starting point for further research that includes other local and international factors in its analysis.

3.1 Literature review for the Facebook Prophet method

The FB Prophet method is a new method that was recently created with the help of data researchers. It has been implemented in several fields, such as tourism, health, financial analysis, urban infrastructure, etc., which has resulted in high accuracy. Breshanaj applied the method successfully in the scientific paper entitled “Modelling and forecasting tourist arrivals in Albania with the FB Prophet model”, published at the International Conference on Innovative Computing and Communication, London, where they created a reliable model for projecting the number of tourists in Albania (Breshanaj, 2024).

A significant analysis was performed on daily COVID-19 infection data in four countries most affected by COVID-19 (USA, Brazil, India, and Russia) to perform a 60-day forecasting of total infections, taking into account live data of the COVID-19 epidemic. This study proposed a prediction method with the FB Prophet model for the analysis of COVID-19. A FB Prophet model is perfect for nonlinear trends that fit daily, weekly, and yearly seasonality plus holiday effects. The FB Prophet model does not require interpolation of missing data, but improves the best forecast from seasonal modeling. Battineni et al., applied this method only to time series data (ds) as model trend terms, allowing space for further research regarding external influencing factors. The proposed model in their work significantly improved estimates of infection numbers in different countries in order to help national authorities (Battineni et al., 2020).

In the study “Enhancing Maritime Safety: A Comprehensive Review of Challenges and Opportunities in the Domestic Ferry Sector” (Baig et al., 2024), the authors addressed an analysis of operational and technological challenges that affect the safety and efficiency of operations in the domestic ferry sector. Maritime safety was the focus of this study, but the authors emphasized that the use of advanced methods, the implementation of supporting policies, and the realization of infrastructure investments to improve maritime operations also play a key role in the efficient management of passenger flows and the minimization of risks. Considering the described challenges by the authors, the application of the advanced method, FB Prophet, for accurate forecasting of passenger optimizes port capacity management and contributes to the development of strategic solutions for similar challenges addressed in this paper.

Traffic flow prediction helps travelers to make better route choices and, thus, reduces travel times and delays. Pragalathan and Schramm created the paper titled “Comparison of SARIMA, Fb-Prophet and Neural Prophet Models for Traffic Flow Predictions at a Busy Urban Intersection”, which presented several models that predicted traffic flow at a busy urban intersection in Duisburg, Germany. After the recent FB Prophet and Neural Prophet models, they checked their performance by showing the metrics (Pragalathan & Schramm, 2024).

Environmental, Social, and Governance (ESG) are the foundation for sustainable and successful operations of various corporations, and sustainability reporting has received much-deserved attention recently. The study “Does Sustainability Reporting Affect Firm Performance? Evidence from the Port Sector” (Gavalas, 2024) analyzed the relationship between transparency in ESG reporting and the financial performance of firms. Based on a sample of 213 publicly listed ports over 5 years, this paper clearly showed that sustainability reporting helps improve operational efficiency, financial stability, and market perception.

This study is a valuable source to support the idea that a commitment to sustainable practices and innovative analytical methods can help optimize port resources and efficiently manage passenger flows. The focus on ESG reporting is closely linked to using FB Prophet to improve forecast accuracy and create strategies that support sustainability objectives. In both the Gavalas study and the present application, transparency and innovative approaches are key to improving port performance and sustainability.

Based on this paper, the application of the FB Prophet methodology for modeling and forecasting port flows can help authorities to develop sustainable strategies in the context of fulfilling ESG objectives to ensure long-term success.

An improved model of Facebook Prophet, with additional layers of positional coding, was developed by Shakeel et al. (2023) in their article “Load forecasting of district heating system based on improved FB-Prophet model” to predict DHN heat consumption. They reviewed and analyzed univariate and multivariate FB-Prophet models. Historical heat load, outdoor temperature, relative humidity, wind speed, wind direction, and weather type of a DHN in Serbia were used to extensively investigate the effectiveness of the improved FB Prophet model. The results showed that the improved FB Prophet model can generate the most accurate and stable forecasts and showed better results for sparse DHN data. The prediction curve fitted the changing trend of hourly DHN consumption, which can effectively guide heat dissipation.

Kumar and Pande (2021) proposed an FB Prophet tool for forecasting supermarket data sales in their research work. They have also examined several forecasting models, such as the additive and the Autoregressive Integrated Moving Average (ARIMA) models, and concluded that the Prophet model is a better forecasting model in terms of low error, better forecasting, and best fit.

The article “Forecasting container throughput of major Asian ports using the Prophet and hybrid time series models” by Munim et al. (2023) explored the use of Prophet and hybrid models to address the complex challenges of forecasting cargo volumes at major Asian ports. Combining Prophet with traditional approaches, such as SARIMA and Holt-Winters, has proven successful in increasing the accuracy of both long-term and short-term forecasts. These findings provide an important precedent for future work on applying Prophet hybrid models to regional ports such as Vlora.

In recent years, investors, corporations, and enterprises have shown a great interest in the Bitcoin network, influencing the growth of scientific research on this topic. The scientific research “Forecasting Bitcoin prices using artificial intelligence: Combination of ML, SARIMA, and Facebook Prophet models” by Cheng, Tiwari, Khaled, Mahendru, and Shahzad used an empirical analysis of financial time series and machine learning to perform Garman-Klass (GK) volatility and Bitcoin price forecasting using short-term memory (LSTM), seasonal autoregressive integrated moving average (SARIMA), and models of the Facebook Prophet. The performance findings showed that LSTM augmentation has a significant improvement over SARIMA and Facebook Prophet in

terms of MSE and MAE. The finding confirmed that Bitcoin values are volatile seasonally, and are often affected randomly by external variables such as cryptocurrency laws, investments, or social media news. Moreover, the results showed a strong optimistic trend, and the days when most people travel are Monday and Saturday, as well as an annual seasonality. SARIMA and FB Prophet models were the best methods for this study (Cheng et al., 2024).

An innovative approach to forecasting flows in ports and transport markets is presented in the paper “Forecasting container freight rates using the Prophet forecasting method” by Saeed et al. (2023), using the Prophet algorithm combined with external factors such as congestion, peak demand, policy, price up, overcapacity, and coronavirus. The main objective of the authors was to improve the accuracy of forecasting transport rates by comparing the performance of models with and without external factors. In this research, it was observed that the inclusion of factors such as overcapacity and coronavirus significantly improved the accuracy of the forecast in the maritime sector. Similarly, the present study uses the Prophet algorithm in modeling passenger flows at Vlora seaport, comparing models with and without the influence of external factors, where it is noted that the inclusion of the COVID-19 factor greatly improves the accuracy of the model.

The FB Prophet method has proven its efficiency and flexibility in its application in various fields. The ability of this method to cope with nonlinear trends, complex seasonality, and missing data makes it the right choice for time series analysis, data modeling, and forecasting. The scientific works reviewed in this chapter show that FB Prophet can contribute to making appropriate decisions and plans in various sectors.

A considerable number of studies have been published on the analysis of flows in international ports or large-scale ports, but advanced studies on regional ports, which play a key role in the development of a country or in international relations, are lacking. This lack creates a significant gap in the existing literature.

Furthermore, existing studies in the port sector do not address the impact of external factors in their studies, but are based only on historical data. The combination of these factors is important, especially in local ports, which are more sensitive to external factors. The present study aims to fill this gap by using the FB Prophet method to model and forecast Vlora seaport, a regional port of strategic importance for Albania and the Mediterranean. While the study focuses on Vlora seaport, the methodology and results of this study can be applied to other ports in the region, providing a starting point for comparison. This study is necessary and has a particular importance in significant infrastructural developments expected to occur in Vlora in 2025. The resizing of the current port into a modern tourist port, “Marina e Vlorës,” and the construction of the first airport in the city, will strengthen Vlora’s position as a major tourist destination in Albania and beyond.

In Albania, there is a lack of scientific works that focus on the forecasting and management of port flows, so this work will create a strong basis for future studies on the coordination and management of passenger flows, by providing an advanced and reliable model which will meet the needs of future infrastructural developments in Vlora.

4. Modeling and forecasting the flow of passengers

To fulfil the purpose of this article, several methods were tested to create a model with high accuracy and reliability for modeling and forecasting passenger flow at Vlora seaport. To evaluate the performance of the different forecasting models, comparative analyses were performed using the metrics RMSE, MAE, MAPE, and the coefficient of fit (R^2). **Table 2** shows the accuracy of each used model, including ARIMA, Holt-Winters, Monte Carlo, ANN, and the FB Prophet model. The results show that the FB Prophet model achieved lower error values (RMSE = 1,356, MAE = 1,364, MAPE = 24.95), and an R^2 coefficient of 0.967, suggesting that it performed better compared to the other models.

The table, with the metrics results, gives a clear picture of the performance of the chosen methods for passenger flow forecasting at Vlora seaport. It is observed that the models based on

traditional statistical approaches, such as Holt-Winters, Monte Carlo, etc., produced relatively high error values, indicating that they were not able to capture the seasonal complexity and structural discontinuities of the present data and are, therefore, not the right models for the study. On the other hand, the artificial neural network (ANN) model, although able to handle non-linear complications, gave a poor performance, possibly because of the deviated figures due to the COVID 19 period. Within the group of models evident as the best, the comparison between the Prophet and ARIMA models is worth noting. Although both models achieved high approximate values for the statistical metrics of model evaluation, Facebook Prophet provided a higher forecast accuracy, as indicated by lower RMSE and MAPE. The data presented in the table clearly indicate that the FB Prophet model not only provides lower error values, but also offers important methodological and practical advantages compared to the ARIMA model. While ARIMA requires stationarity of the data, the Prophet method automatically handles trend, seasonality, and unexpected changes through a flexible additive structure. Another advantage of the Prophet method is that it allows the inclusion of macroeconomic factors, such as GDP and exchange rate, in a natural way in the model, which makes it a valuable tool for strategic decision-making. In this context, Prophet's structural advantages and higher performance make it significantly more suitable for modeling data such as passenger flow at Vlora seaport, data that is characterized by seasonal fluctuations, external influences, and disruptions such as those caused by the COVID-19 pandemic.

Table 2 Comparison of the performance of different models applied for passenger flow modeling and forecasting.

Model	RMSE	MAE	MAPE	R2	Features
ARIMA	2,197	1,472	32.89	0.96	A linear model that captures statistical time dependence.
Holt-Winters	5,493	3,170	45.84	0.75	An exponential smoothing model suitable for data with trend and seasonality.
Monte Carlo	5,453	3,138	45.35	0.753	A probabilistic simulation method for forecasting with possible value distributions.
Artificial Neural Network	7,872	4,454	49.34	0.69	A nonlinear model that captures complex structures through learned weights.
Facebook Prophet	1,356	1,364	24.95	0.967	A flexible model for time series data with trend and seasonality designed for forecasting.

4.1 Modeling the passenger flow in Vlora seaport

Facebook Prophet was created by Facebook's Core Data Science Team in 2017 as a powerful method for modeling and analyzing time series, from the simplest to the most complex seasonal time series. This method was created to help many economic researchers, data researchers, and scientists in different fields who want to make the highest accuracy modeling and predictions in their projects.

The FB Prophet model is designed to be easy to use, with a simple syntax for building and using forecast models. It uses a methodology with a hybrid approach, combining elements of additive and multiplicative time series models, including seasonality, trend, and sensitive external changes. Combining the above elements, the structure of the model is:

$$y_n = g(t) + s(t) + h(t) + \varepsilon_t \text{ where :}$$

$y(t)$ represents the time series that are forecasted at time t ;

$g(t)$ represents the time series trend;

$s(t)$ represents seasonality;

$h(t)$ represents the external factors that affect the time series.

ε_t represents the difference between the fundamental values and those predicted by the model.

FB Prophet can be adapted to much different software by modeling and predicting data that have missing data, extreme values, or different trends. In many tourism time series projects, the FB Prophet method has been quite successful compared to other methods used due to the seasonal nature of the tourism series. This method works naturally for seasonal data with different seasons, such as hourly, daily, weekly, or yearly. The Prophet method is capable of handling data that change the trend over time or change it immediately (change points), as in the case of the drastic change in trend of the number of arrival passengers at Vlora seaport due to the COVID-19 pandemic.

Another feature that makes this method effective for the study is the given possibility by FB Prophet to interact with the influence of external factors, such as the exchange rate, GDP, or the COVID-19 pandemic, in the modeling and forecasting of passenger flow in Vlora seaport.

The *plotly* package in R software was used to present historical data as an interactive graph. This powerful and flexible tool for creating advanced graphs and visualizations allowed further exploration of data and graphs.

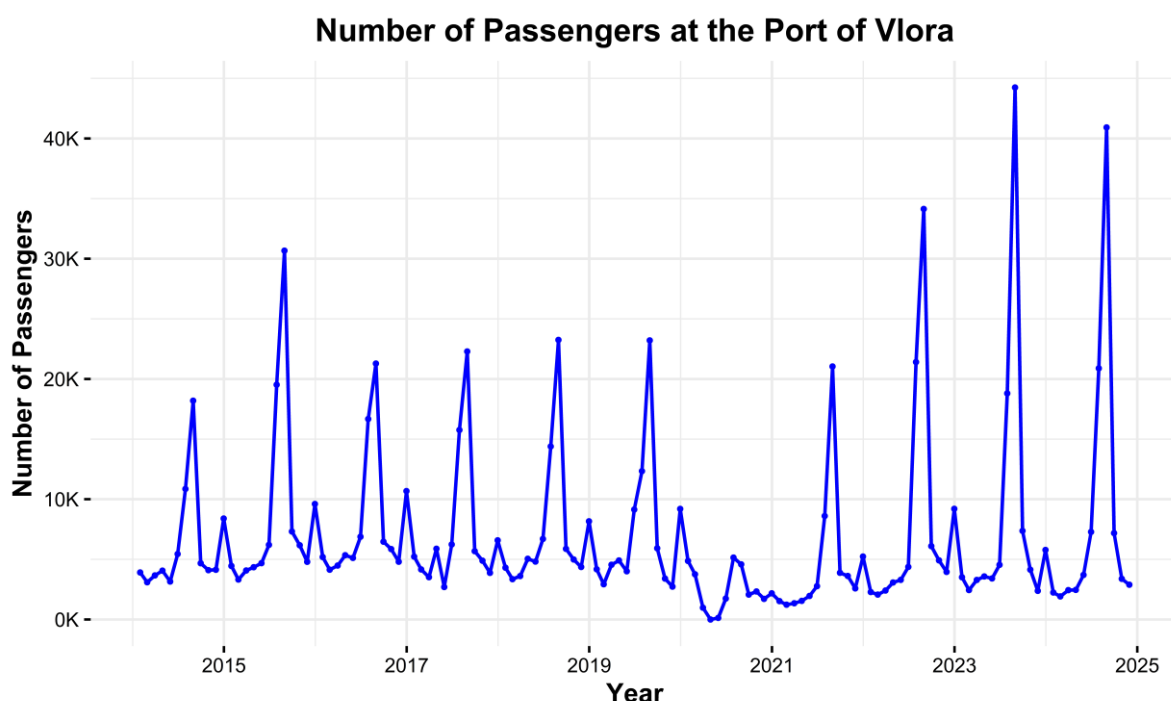


Figure 5 Number of passengers at Vlora seaport (<https://rpubs.com/Breshanaj/1254925>).

In **Figure 5**, it is seen that the data recorded for the arrival passengers at the Vlora seaport had a slight upward trend for the years 2014 - 2020 but, in 2020, it had a drastic decrease, where the month of April 2020 recorded the minimum value; zero passengers arrived after work at the seaport as it was suspended due to the Covid19 pandemic, but then the number of arrivals passengers began to increase gradually, surpassing the values of arrivals before the pandemic, recording the maximum value of 44,242 arriving passengers at Vlora seaport for August 2023.

For the modeling and forecasting of the data for the incoming flow at Vlora seaport, the “*prophet*” package in Rstudio was used, which provided the following information:

The model predicted a linear increase in the number of incoming passengers at the Vlora seaport.

The model identified 25 change points, such as “2016-04-30” and “2020-08-31,” that indicated major changes immediately in the data trend.

The annual seasonality was active, with a period of 365.25 days related to the tourist seasonality.

The model created a confidence interval using 1,000 scenarios to ensure that future values varied within a certain range. Main parameters:

K: 0.0655 (moderate increase forecast)

M: 0.1919 (long-term change)

Sigma_obs: 0.078 indicates that the error is relatively small, compared to the scale of the data.

The metrics R^2 , $RMSE$, $MAPE$ and MAE were calculated to evaluate the model's performance. The value $R^2 = 0.967$ (this value is 0.98 for the period 2020 - 2024) showed a good fit for the model, explaining about 97 % of the variation in the data.

The “predict” code in Rstudio was used to make a forecast for the next few years and present the result through the interactive graph in **Figure 6**. The seasonal nature and the growing trend for the next three years are clearly visible.

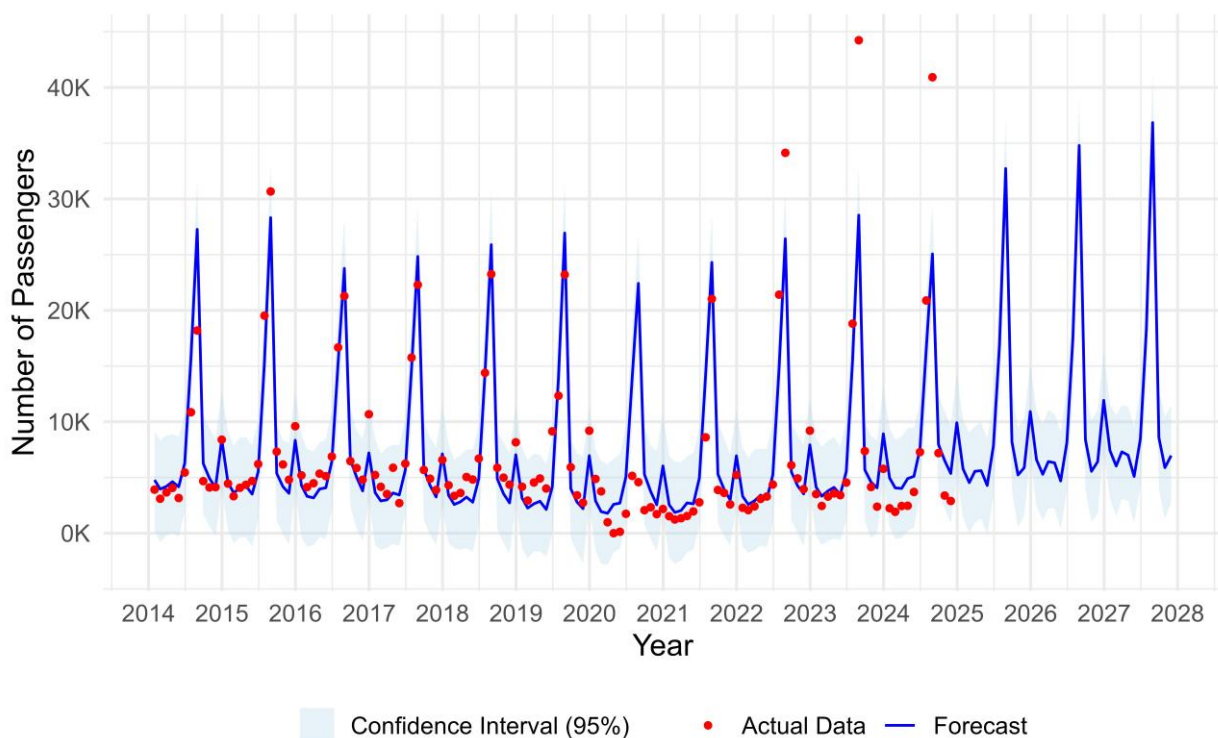


Figure 6 Forecast graph for three years (<https://rpubs.com/Breshanaj/1254926>).

According to this model, the minimum value will be reached in May 2025, with 4,285 passengers, and the maximum value of 37,000 passengers is expected to be registered in August 2027.

To make this model even more effective, data from 2020 (and earlier) that corresponded with a pandemic period (COVID-19) were removed. The Facebook Prophet for the period 2021 - 2024 was remodeled. This model resulted in a very high accuracy, represented by the metrics $R^2 = 0.98$ and $MAPE$ 23, ensuring that the predictions of future values will be close to the real values. This significant difference between the two above scenarios occurred mainly due to the COVID-19 pandemic.

Figure 7 presents the modeling of historical data using this algorithm. The high value of the adaptation coefficient shows the model's high accuracy in adapting the existing data. Based on this high performance, the model is used to make predictions on the passenger flow in the Vlora seaport for the period 2025 - 2028, providing a reliable and stable analysis for future trends.

The figure clearly shows the increasing trend of passenger flow in Vlora seaport. It is predicted that the highest number of passengers will be reached in August 2027, with about 49,000.

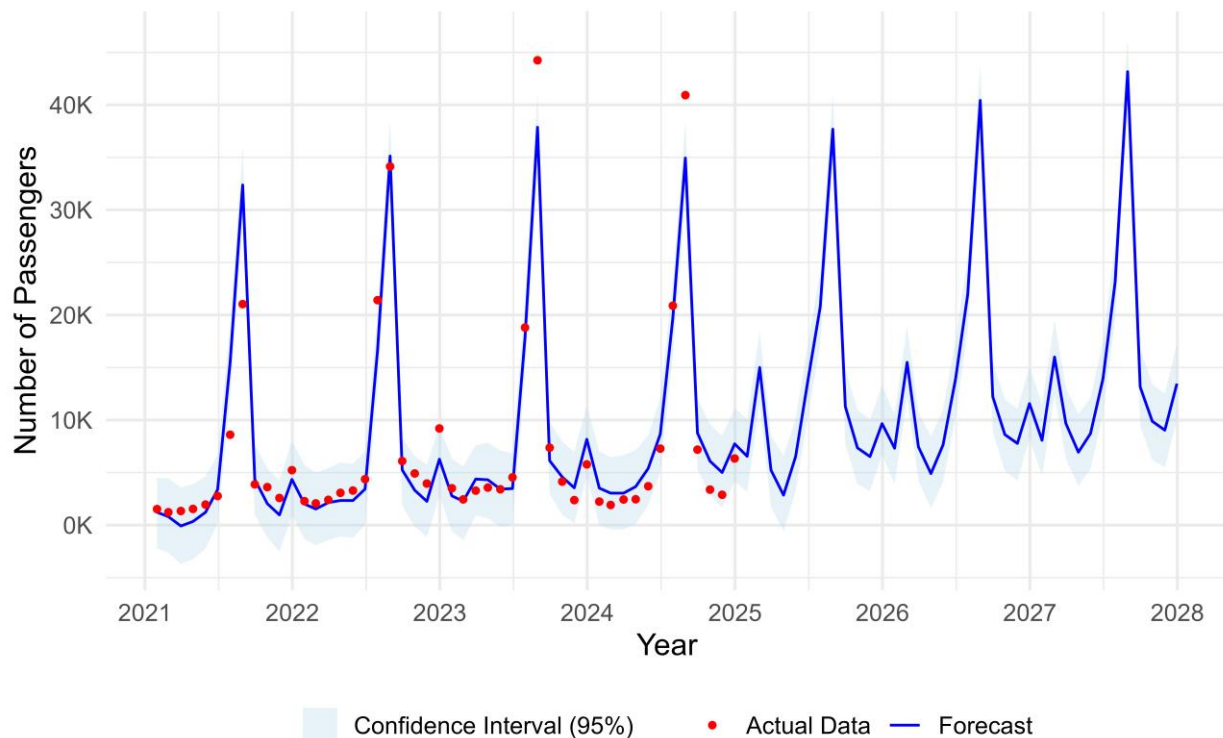


Figure 7 Forecast graph for 2025 - 2028 (<https://rpubs.com/Breshanaj/1254934>).

Using cross-validation, Luiz et al. (2024) the FB Prophet2 model achieved the following metrics;

Mean Absolute Error (MAE): 1,249.158

Root Mean Squared Error (RMSE): 1,322.529

Mean Absolute Percentage Error (MAPE): 23

R^2 (Coefficient of Determination): 0.98

which show an improvement in the above model. This improvement reflects the impact of the COVID-19 factor on effectiveness. The impact of other factors, such as GDP and the euro-lek exchange rate, is studied in subsection 4.2.

4.2 Modeling the passenger flow in Vlora seaport under the influence of external factors

In this section, FB Prophet is used to model and predict the passenger number in Vlora seaport, taking into account not only historical passengers data, but also macroeconomic data, such as the exchange rate or the Gross Domestic Product, which impact the issue studied as external factors.

The exchange rate can affect the costs and preferences of passengers, especially international ones, encouraging or prohibiting passenger flow according to the strength or weakness of the local currency.

The average exchange rate of foreign currencies against the lek represents the simple average of the official daily exchange rates against the local currency lek. The official exchange rate is calculated according to the Instruction “On the method of calculating the “fixed” exchange rate of the lek, “approved by the decision of the Supervisory Council, No. 66, dated 30.08.2006 (Albanian Bank, 2024).

GDP, an indicator of a country's economic health, can directly affect the number of passengers travelling for tourism, business, family visits, etc.

According to official World Bank data, Albania's Gross Domestic Product (GDP) was 22.98 billion US dollars in 2023. The value of Albania's GDP represents 0.02 % of the world economy (World Bank, 2024).

In the graph below, all three of these lines are presented in the same graph. The exchange rate values look like a horizontal line because the fluctuations in the exchange rate are very small compared to the number of passengers in Vlora seaport (**Figure 8**).

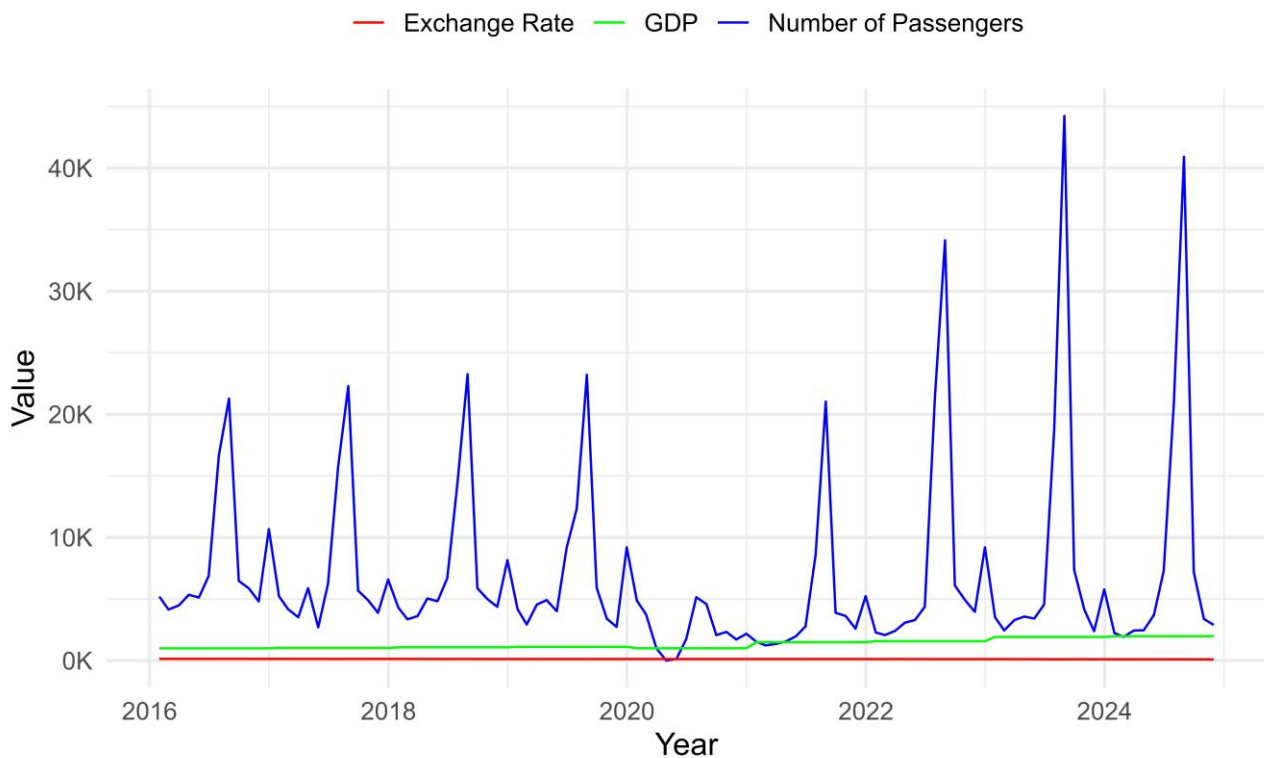


Figure 8 Real values of passenger numbers, exchange rate, and GDP (<https://rpubs.com/Breshanaj/1255032>).

If these are looked at as a separate graph, it is seen that the exchange rate had a strong decline from 2016 to 2024.

In 2016, one euro was exchanged for 140 lek. In 2023, 1 lek was equal to the European currency, and now 1 euro is 98 Albanian lek. The GDP has doubled in value, from 11.86 billion USD in 2016 to nearly 24 billion USD in 2024.

Modeling and forecasting, according to this method, considering the influence of external factors, confirms once again that the number of arriving passengers at Vlora seaport will continue to show an upward trend. However, the accuracy of this model is moderated, since only 68 % of the variance in the number of passengers can be explained by the used variables (exchange rate, COVID-19 period, and GDP) in the model. The values of the respective correlations between the examined variables also support this result:

The coefficient of correlation between the exchange rate and the data is (-0.12) , which is a weak negative correlation. The too-small value of this indicator supports the fact that this factor has no impact on the number of incoming passengers in Vlora seaport.

The correlation between GDP and y is 0.046, a weak positive correlation. This correlation shows that an increase in GDP would increase the number of passengers, but this growth would be negligible in terms of its value.

The correlation between the exchange rate and GDP is (-0.93) , a strong negative correlation, which shows that the decrease in the euro-lek exchange rate (valuation of the Albanian currency) leads to an escalation in the magnitude of the GDP.

According to this model, the number of arriving passengers at the port will continue to increase, with the maximal number reached in August 2026, with 32,600 incoming passengers.

Figures 9 and 10 reflect their graphic presentation.

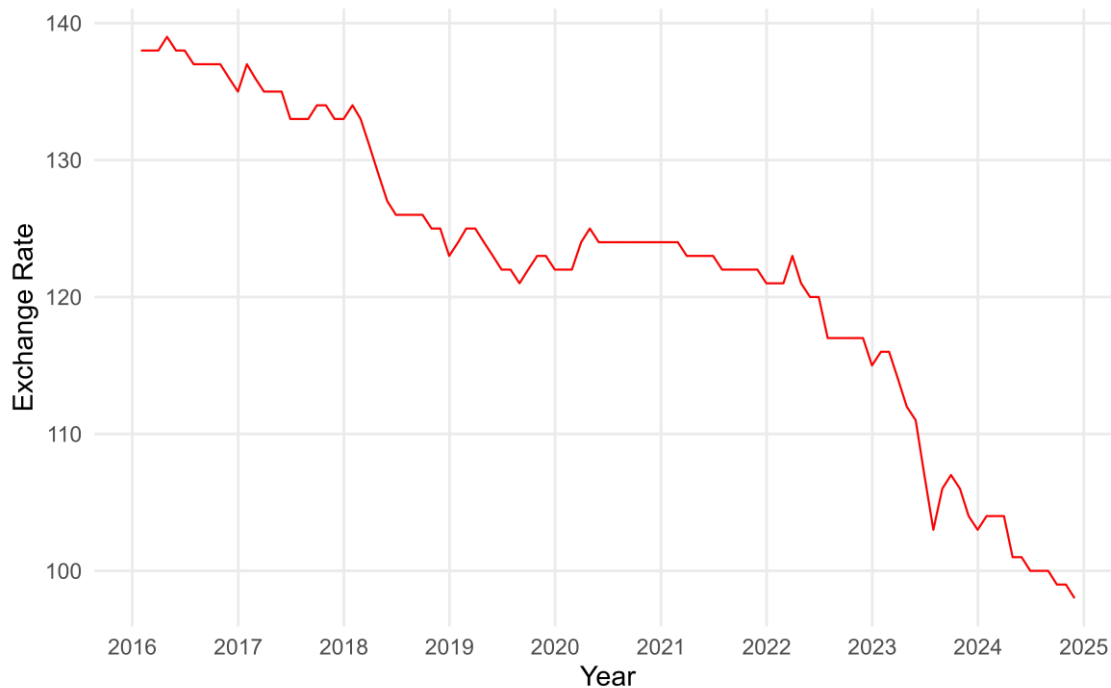


Figure 9 Euro-lek exchange rate 2016 - 2024.

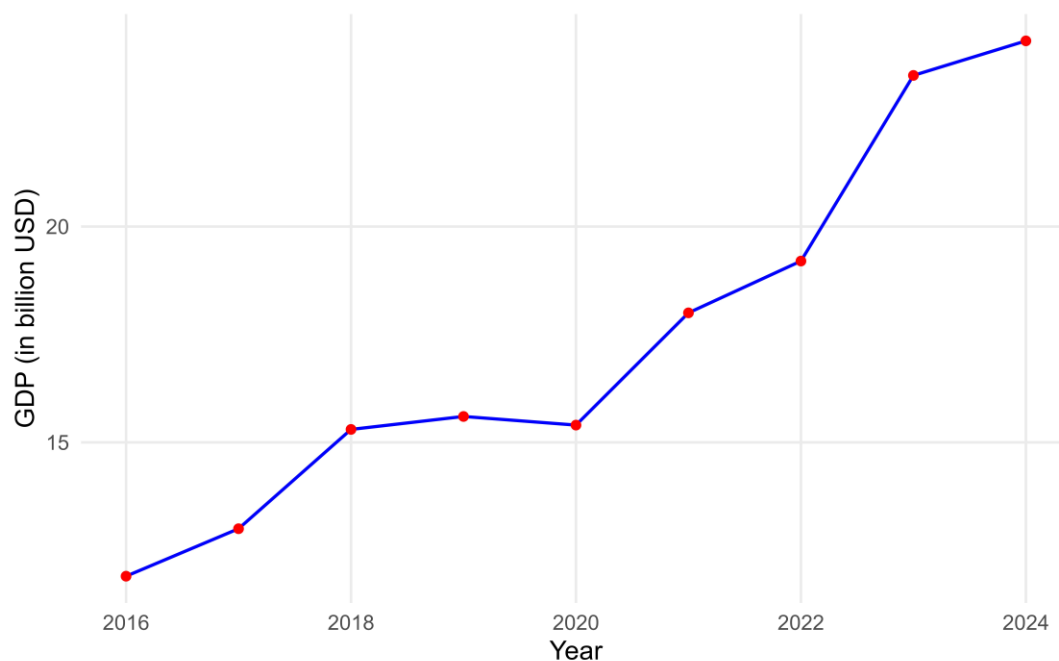


Figure 10 Development of GDP in 2016 - 2024.

The interactive graph below reflects the passenger flow forecast in Vlora seaport (**Figure 11**).

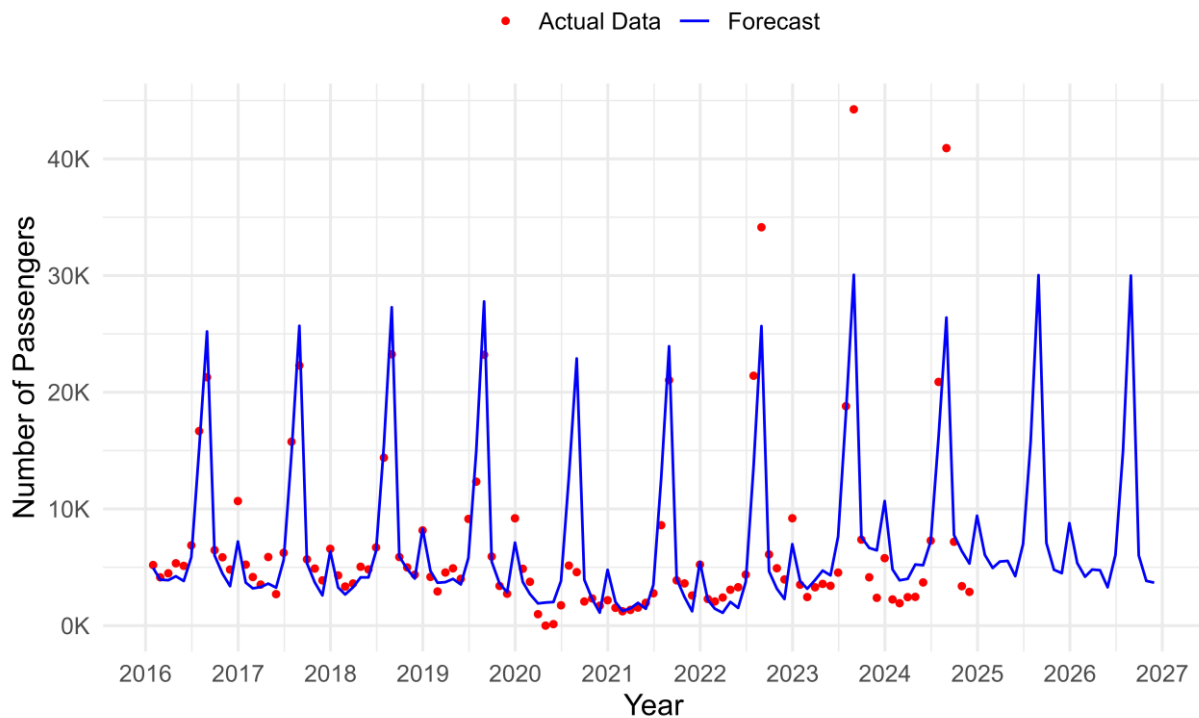


Figure 11 Graphic of passengers forecast with the impact of external factors (<https://rpubs.com/Breshanaj/1255078>).

The Facebook Prophet model was adapted for the period 2021 - 2024, considering the influence of external factors, such as the exchange rate and GDP. As was expected, this modeling is more accurate than the above model, confirming once again that the data for the period of the COVID-19 pandemic deforms the model and reduces its accuracy.

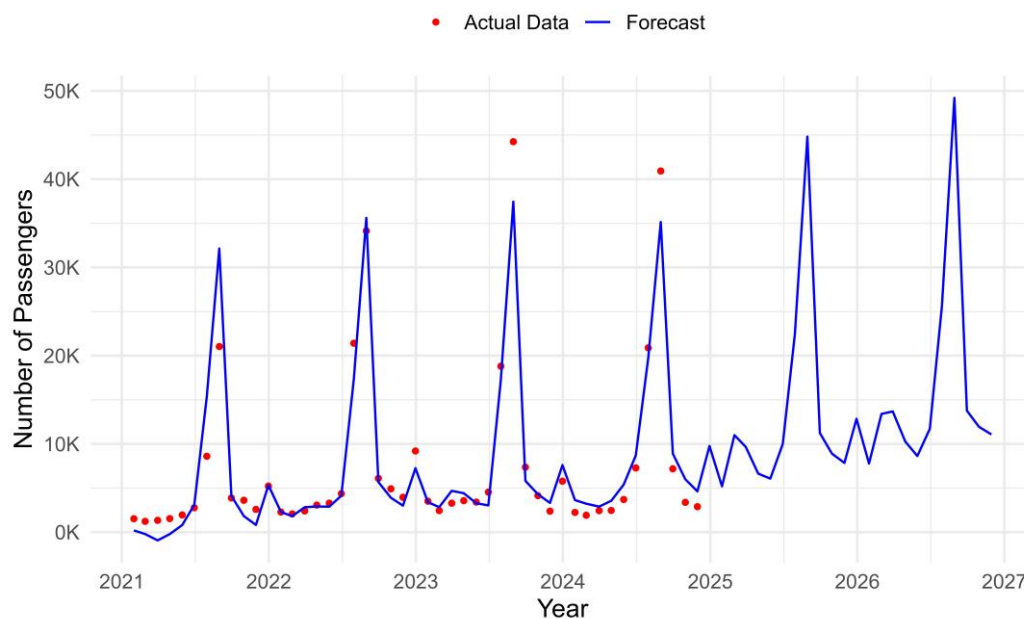


Figure 12 Graphic of passengers forecast with the impact of external factors from 2021 (<https://rpubs.com/Breshanaj/1255080>).

In **Figure 12**, the results of this model are graphically represented, highlighting the growth trend, seasonality, modeling, and forecasting of the number of passengers entering the port under the influence of external factors, such as GDP and exchange rate, after removing the COVID-19 factor.

These results demonstrate the suitability of the FB Prophet model in the interaction of historical data with macroeconomic variables to build a valid modeling and forecast that helps in the development of infrastructure in the region.

5. Results

This study followed a systematic process beginning with the collection and cleaning of historical passenger flow data from 2014 to 2024, followed by the integration of external factors such as GDP and exchange rate. Next, various forecasting models (including ARIMA, Holt-Winters, Monte Carlo, and Artificial Neural Networks) were applied and evaluated according to standard accuracy metrics. Analysis revealed that the Facebook Prophet model, when applied only to historical data for the period 2021 - 2024, exhibited excellent performance, with an R^2 coefficient of 0.98 and a MAPE of 24.95 %, capturing 98 % of the data variance. **Figure 13** schematically presents the main steps taken and the major results of the study, providing a visual summary that makes it easier for the reader to follow the methodological flow and results.

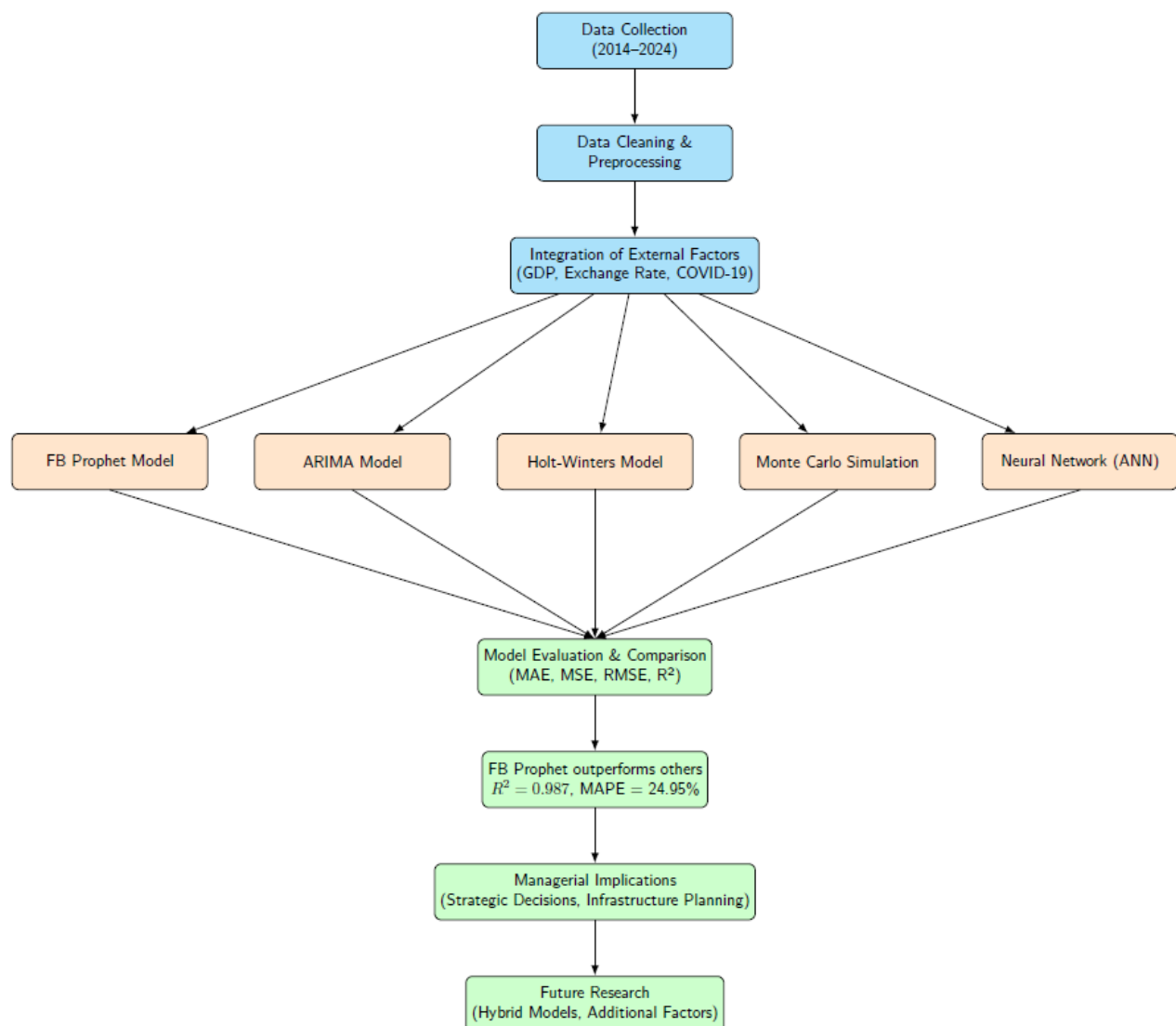


Figure 13 Work scheme.

Since the FB Prophet model has proven to be more efficient than other forecasting methods, it was decided to further improve it by building four different scenarios, which examined the impact of the data period and the integration of external factors. These scenarios allowed the assessment of the accuracy of the model depending on the use of complete historical data (2014 - 2024), compared to those after the Covid19 period (2021 - 2024), as well as the integration of factors such as exchange rate and GDP. Mark:

Prophet1 = the FB Prophet model, which is based only on historical data for the period 2014 - 2024;

Prophet2 = the FB Prophet model, which is based only on historical data for the period 2021 - 2024;

Prophet3 = the FB Prophet model, which is based on historical data and the influence of external factors such as the exchange rate and GDP for the period 2016 - 2024;

Prophet4 = the FB Prophet model, which is based on historical data and the influence of external factors such as the exchange rate and GDP for the period 2021 - 2024.

According to the results of each scenario, **Table 3** was built with the accuracy metrics:

Table 3 Table of metrics for each FB Prophet script.

Model	MAE	RMSE	MAPE	R ²
Prophet1	1,364	1,356	24.95	0.967
Prophet2	1,249.158	1,322.529	23	0.98
Prophet3	2,024.18	3,390.121	42.0829	0.8057466
Prophet4	1,678.757	2,655.787	34.99977	0.9275736

The table shows that all the metrics converge to the same model, confirming that the model with the highest accuracy is FB Prophet2. The main findings of the study show that the FB Prophet model applied only to historical data for the period 2021 - 2024 provided accurate predictions. This model had a good fit and accounted for 98 % of the variance in the data with a mean absolute percentage error (MAPE) of 23 %. In comparison, the models that also included context factors (Prophet3 and Profet4) performed worse than the previous two models. All models applied, regardless of their accuracy, confirmed that there is a clear seasonal pattern with an increasing trend in passenger flow at Vlora seaport, with peaks in the summer months.

This study provides a good example of how methodological tools can be applied to support the work of local authorities and transport service providers in planning and managing future developments. The results were not very effective when external factors were also integrated. This suggests that other local factors, such as climate, population, ferry ticket prices, and the opening of Vlora airport may be more influential factors in modeling the inflow into Vlora seaport, which offers opportunities for further research. Furthermore, the analyses suggest that, in the future, the effects of the interaction of air-sea transport should be re-evaluated after the reconstruction of the port and the opening of the Vlora airport. In conclusion, the application of the FB Prophet model for predicting passenger flow in Vlora seaport was efficient, and the study also helps to address a missing link in the Albanian literature on this topic and, consequently, offers practical recommendations for the daily and strategic management of the port in a rapidly developing economic and urban environment. In addition to general managerial suggestions related to the number and type of service lines or prices, this analysis provides some concrete recommendations that can help improve the practical functioning of Vlora seaport. First, due to the expected increase in passenger flow in the summer months, it is recommended to expand and modernize the passenger terminal, adding more efficient waiting and entry-exit spaces. Secondly, the port needs to implement advanced performance

monitoring systems which track passenger processing time and delays and daily loads in real time. Third, improving the quality of service can be achieved through periodic staff training, with a focus on passenger interaction and flow management under peak conditions. These direct interventions, guided by forecasting models such as Prophet, provide a solid basis for strategic planning and sustainable development of the port in line with seasonal and international transport demands.

6. Discussions

Passenger seaports are key points for urban development, and engines for improving infrastructure and increasing the tourist attractiveness of a city and region (Krile et al., 2021).

This study used the FB Prophet method to model and forecast for a regional port by integrating the impact of external factors, such as GDP or exchange rate. The modeling was evaluated by an accuracy metric R^2 of 0.98, a value very close to the perfect value, classifying this method with high effectiveness. These results are consistent with the findings of Shakeel et al. (2023), who reported that the FB Prophet method is more suitable for time series that include external influences and unexpected changes.

Compared to models using only historical data, the inclusion of factors such as GDP and exchange rate in this study resulted in lower accuracy ($R^2 = 0.8057$). This is consistent with the results of Cheng et al. (2024), who found that macroeconomic factors often have a limited impact on forecasts of passenger or freight flows.

The findings on the peaks of flows in July and August are similar to those of Breshanaj (2024) who used FB Prophet to forecast tourist flows in Albania. This reinforces the importance of seasonality in models for regional ports and tourist destinations.

Most existing studies focus on international or large ports (Gavalas, 2024; Baig et al., 2024), while regional ports such as Vlora lack advanced analysis. The present study contributes to this gap by providing a reliable model for regional ports.

In line with recent literature, it is found that pandemic data negatively impacts model accuracy (Battineni et al., 2020). The importance of addressing anomalies caused by global crises is emphasized in order to increase forecast accuracy.

The results are important in assisting local authorities and transport operators in the strategic management of port capacities, especially considering the construction of the tourist port and Vlora airport in 2025. This is consistent with the suggestions of Baig et al. (2024) on the importance of advanced models in managing passenger flows in ports of strategic importance.

The analytical results of this study will enable informed decision-making by port authorities and operators, such as strategic decisions regarding the number and type of service lines, type and size of ships, pricing policies, public service obligations, number of employees, and staff organization, as well as state policies for port infrastructure and their financing. These decisions will influence the optimization of services, increase safety, and improve competitiveness by ensuring better integration into the international transport network.

The predictive system can be generalized to other areas of operations management, such as hospitals or theme parks, in which people flows need to be accurately predicted (Guo et al., 2021). A subsequent study could focus on the interaction between air and sea transport after the opening of Vlora airport in 2025.

7. Conclusions

The FB Prophet method in this research has proven effective in modeling complex trends and prominent seasonality in time series data, with an R^2 of 98 % and a MAPE of 23 %. This result indicates that the proposed model is reliable in its predictions and can be helpful in Vlora seaport's strategic decision-making and operational actions. This research contributes to the Albanian context as the first to present an artificial intelligence model for predicting passenger flow in seaports. However, this study has several limitations, the main one being that it is based primarily on historical

data and partially includes external factors. Although some of these factors included in the analysis did not have a significant effect, this may mean that there are other factors that can increase the accuracy of the models. In the future, more data should be used, other factors should be used, and hybrid models should be applied to increase the accuracy of the models. This study provides an advanced methodological framework to support infrastructure investments and improve port management with the aim of increasing port performance. The findings of this study not only contribute to the connection of Vlora seaport with the international transport system, but provide a reasonable basis for further investigations into the forecasting and management of passenger flows in ports.

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